

Assignment 3 — Write-up

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Part A — A Mini-Course on Concentration Inequalities

This mini-course develops the four concentration inequalities used (without proof) in the Week 3 derivation of the i.i.d. growth-function ERM guarantee:

- Hoeffding’s inequality,
- Hoeffding’s inequality for sampling without replacement,
- McDiarmid’s bounded-differences inequality,
- Bernstein’s inequality.

The exposition is self-contained: every result we cite is proved here. We close by gluing the pieces together to give a full proof of the Week 3 ERM bound.

A.1 The concept of concentration

What concentrates. A *concentration inequality* is a quantitative statement that some random quantity Z — typically a function $Z = f(X_1, \dots, X_n)$ of many independent inputs — is, with high probability, very close to a deterministic value (usually its mean $\mathbb{E}Z$, sometimes its median). Concretely, we want bounds of the form

$$\Pr(|Z - \mathbb{E}Z| \geq t) \leq 2 \exp(-ct^2/v),$$

where v is some “scale” parameter (a sum of squared ranges, a variance proxy, etc.). The quadratic-in- t exponent is what we mean by *sub-Gaussian* tails.

Why we should expect this. The intuition is the Law of Large Numbers turned quantitative. For $S_n = \frac{1}{n} \sum_{i=1}^n X_i$ with i.i.d. bounded X_i , the variance of S_n is σ^2/n , so the standard deviation is $O(1/\sqrt{n})$ — much smaller than the range of an individual X_i . The CLT says the *shape* of the fluctuations is Gaussian in the limit; concentration inequalities give *non-asymptotic* Gaussian-like tails for finite n . The phenomenon extends beyond sums to any function of $X_1, \dots, X_n$ that depends only mildly on each coordinate (this is the content of McDiarmid’s inequality).

Concrete example. Toss a fair coin n times and let S_n be the number of heads. Markov’s inequality on $|S_n - n/2|$ gives no useful bound. Chebyshev’s inequality, using $\text{Var}(S_n) = n/4$, gives $\Pr(|S_n - n/2| \geq t\sqrt{n}) \leq 1/(4t^2)$ — polynomial decay only. Hoeffding’s inequality (proved below) gives the much sharper $\Pr(|S_n - n/2| \geq t\sqrt{n}) \leq 2 \exp(-2t^2)$ — exponential in t^2 . For $n = 10000$ and $t = 3$, Chebyshev says “ ≤ 0.028 ”; Hoeffding says “ $\leq 2.5 \cdot 10^{-8}$ ”. This gap is typical and is why concentration inequalities are the right tool for learning theory: generalization guarantees need failure probability $\delta \rightarrow 0$, not just $\delta \leq 0.5$.

The intuition behind concentration is *averaging cancels independent noise*. Independence prevents the deviations from one coordinate piling up systematically with the deviations from another; bounded influence of each coordinate prevents any single coordinate from dominating; together, they force the whole quantity to look like its mean.

A.2 The common technique: the MGF / Chernoff method

The proofs of Hoeffding, Bernstein, and (with a martingale twist) McDiarmid all share one backbone: **bound the moment generating function, then optimize via Chernoff.**

A.2.1 The Chernoff template

Let Z be any real random variable and $\lambda > 0$. By Markov's inequality applied to the non-negative random variable $e^{\lambda Z}$,

$$\Pr(Z \geq t) = \Pr(e^{\lambda Z} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E} e^{\lambda Z}. \quad (\text{Chernoff})$$

Optimizing over $\lambda > 0$,

$$\Pr(Z \geq t) \leq \exp\left(-\sup_{\lambda > 0} [\lambda t - \log \mathbb{E} e^{\lambda Z}]\right).$$

The function $\psi_Z(\lambda) := \log \mathbb{E} e^{\lambda Z}$ is the *cumulant generating function* (CGF), and the supremum above is the Legendre dual $\psi_Z^*(t)$.

The job, then, is to upper bound $\psi_Z(\lambda)$. The bigger your upper bound on ψ_Z , the smaller the dual ψ_Z^* , and the worse the concentration. For sums of independent variables this is easy: independence converts the MGF of a sum into a product, hence the CGF of a sum into a sum of CGFs:

$$Z = \sum_{i=1}^n X_i \implies \psi_Z(\lambda) = \sum_{i=1}^n \psi_{X_i}(\lambda).$$

So we only need to bound the CGF of each individual X_i . Two standard routes:

- **Hoeffding's lemma:** bounded random variables have $\psi_X(\lambda) \leq \lambda^2(b-a)^2/8$. This gives sub-Gaussian tails depending only on the *range* of X .
- **Bernstein's lemma:** variance-aware bound that yields tighter tails when $\text{Var}(X) \ll (b-a)^2$.

A.2.2 The bounded-differences variant (preview of McDiarmid)

For functions $Z = f(X_1, \dots, X_n)$ that are not literal sums, independence still helps, but we need a martingale decomposition. Let $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$ and $D_k := \mathbb{E}[Z \mid \mathcal{F}_k] - \mathbb{E}[Z \mid \mathcal{F}_{k-1}]$. Then

$$Z - \mathbb{E}Z = \sum_{k=1}^n D_k,$$

and (D_k) is a martingale difference sequence. If we can bound the conditional MGF of each D_k by a sub-Gaussian term, the same Chernoff machinery goes through. Section A.4.3 makes this precise.

A.3 Statements, assumptions, and consequences

In all four statements, $X_1, \dots, X_n$ are independent (with one exception, noted) and $t > 0$.

A.3.1 Hoeffding's inequality

Theorem (Hoeffding, 1963). Let $X_1, \dots, X_n$ be independent with $X_i \in [a_i, b_i]$ almost surely. Let $S = \sum_{i=1}^n X_i$. Then

$$\Pr(S - \mathbb{E}S \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and the same bound for the lower tail; a union bound gives the two-sided version with a factor of 2.
Assumption: boundedness and independence. No assumption on means or variances.

For the i.i.d. mean $\bar{X} = \frac{1}{n} \sum X_i$ with $X_i \in [a, b]$:

$$\Pr(|\bar{X} - \mathbb{E}\bar{X}| \geq \varepsilon) \leq 2 \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

A.3.2 Hoeffding's inequality for sampling without replacement

Theorem (Hoeffding, 1963; Serfling, 1974). Let $\mathcal{C} = \{c_1, \dots, c_N\} \subset [a, b]$ be a finite population with mean $\mu = \frac{1}{N} \sum c_i$. Draw $X_1, \dots, X_n$ uniformly *without replacement* from $\mathcal{C}$, $n \leq N$. Then

$$\Pr\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \geq \varepsilon\right) \leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

Assumption: uniform sampling without replacement from a bounded population. **No independence:** the X_i are exchangeable but dependent. Strikingly, the bound is *at least as good* as the with-replacement Hoeffding bound — sampling without replacement concentrates *more*, because exhausting the population reduces variability. This is what the Week 3 proof needs at the symmetrization step (a ghost sample is drawn without replacement from a fixed pool).

A.3.3 McDiarmid's inequality (bounded differences)

Theorem (McDiarmid, 1989). Let $X_1, \dots, X_n$ be independent random variables on spaces $\mathcal{X}_1, \dots, \mathcal{X}_n$, and let $f : \prod \mathcal{X}_i \rightarrow \mathbb{R}$ satisfy the *bounded-differences* property: there exist constants $c_1, \dots, c_n \geq 0$ such that for every i and every $x_1, \dots, x_n, x'_i$,

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

Then

$$\Pr(f(X) - \mathbb{E}f(X) \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Assumption: independence + bounded coordinate-wise influence. No structural assumption on f (no convexity, no Lipschitz in any norm beyond the discrete bounded-differences condition). When $f(x) = \sum_i x_i/n$ and $x_i \in [a, b]$, $c_i = (b-a)/n$, and we recover Hoeffding (up to the factor of n in the rescaling). McDiarmid's strength is that it applies to any bounded-influence function; in learning theory, this is what controls the uniform deviation $\sup_{h \in \mathcal{H}} |L_{\mathcal{D}}(h) - L_n(h)|$.

A.3.4 Bernstein's inequality

Theorem (Bernstein). Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = 0$, $|X_i| \leq M$ a.s., and $\sum_i \mathbb{E}X_i^2 \leq v$. Then for all $t \geq 0$,

$$\Pr\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{t^2}{2(v + Mt/3)}\right).$$

Equivalently, with probability at least $1 - \delta$,

$$\sum X_i \leq \sqrt{2v \log(1/\delta)} + \frac{2M}{3} \log(1/\delta).$$

Assumption: independence, boundedness, and a variance bound v . **What Bernstein exploits beyond Hoeffding:** the *variance* of each X_i , not just its range. When $\text{Var}(X_i)$ is much smaller than M^2 (rare events, low-probability indicators, Bernoulli with p small), the bound is dramatically sharper: in the small-deviation regime $t \ll v/M$ Bernstein gives Gaussian tails with the *true* variance proxy v rather than the loose proxy nM^2 that Hoeffding uses.

A.3.5 Side-by-side comparison

Inequality	Quantity	Assumption	Tail at level t
Hoeffding	$\sum(X_i - \mathbb{E}X_i)$, indep.	$X_i \in [a_i, b_i]$	$\exp(-2t^2 / \sum(b_i - a_i)^2)$
Hoeffding (w/o replacement)	$\sum X_i - n\mu$ from finite pop.	$c_i \in [a, b]$, sampled w/o repl.	$\exp(-2t^2 / (n(b - a)^2))$
McDiarmid	$f(X) - \mathbb{E}f(X)$	indep. + bdd. diffs. c_i	$\exp(-2t^2 / \sum c_i^2)$
Bernstein	$\sum X_i$, indep., zero-mean	$ X_i \leq M$, $\text{var} \leq v$	$\exp(-t^2 / (2v + 2Mt/3))$

The first three are *purely sub-Gaussian*: their exponent is quadratic in t with a scale that uses only ranges/diameters. Bernstein is *sub-exponential beyond a threshold*: in the small- t regime it beats Hoeffding by replacing $\sum(b_i - a_i)^2/4$ with the typically much smaller v ; in the large- t regime ($t \gtrsim v/M$) it transitions to a slower $\exp(-3t/(2M))$ tail, which is the price for the variance-aware constant.

A.4 Proofs

A.4.1 Hoeffding's lemma and Hoeffding's inequality

Lemma (Hoeffding's lemma). If $X \in [a, b]$ a.s. and $\mathbb{E}X = 0$, then for all $\lambda \in \mathbb{R}$,

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Proof. By convexity of $x \mapsto e^{\lambda x}$, for $x \in [a, b]$,

$$e^{\lambda x} \leq \frac{b-x}{b-a}e^{\lambda a} + \frac{x-a}{b-a}e^{\lambda b}.$$

Taking expectations and using $\mathbb{E}X = 0$,

$$\mathbb{E}e^{\lambda X} \leq \frac{b}{b-a}e^{\lambda a} - \frac{a}{b-a}e^{\lambda b} = e^{\phi(\lambda)},$$

where, setting $p = -a/(b-a) \in [0, 1]$ and $u = \lambda(b-a)$,

$$\phi(\lambda) = -pu + \log(1 - p + pe^u).$$

Direct computation gives $\phi(0) = \phi'(0) = 0$ and $\phi''(u) = \frac{p(1-p)e^u}{(1-p+pe^u)^2} \leq \frac{1}{4}$ (since the function $q(1-q)$ with $q = pe^u/(1-p+pe^u) \in [0, 1]$ is at most $1/4$). Taylor's theorem gives $\phi(\lambda) \leq \lambda^2(b-a)^2/8$. $\square$

Hoeffding's inequality (proof). WLOG centering each X_i (replace X_i by $X_i - \mathbb{E}X_i \in [a_i - \mathbb{E}X_i, b_i - \mathbb{E}X_i]$, an interval of length $b_i - a_i$). Let $Z = \sum(X_i - \mathbb{E}X_i)$. By independence and Hoeffding's lemma,

$$\mathbb{E}e^{\lambda Z} = \prod_i \mathbb{E}e^{\lambda(X_i - \mathbb{E}X_i)} \leq \prod_i e^{\lambda^2(b_i - a_i)^2/8} = e^{\lambda^2 V/8}, \quad V := \sum(b_i - a_i)^2.$$

By Chernoff, $\Pr(Z \geq t) \leq \exp(-\lambda t + \lambda^2 V/8)$, minimized at $\lambda^* = 4t/V$, giving $\exp(-2t^2/V)$. $\square$

A.4.2 Hoeffding without replacement

We prove the result via Hoeffding's elegant *coupling reduction* to the i.i.d. case: sampling without replacement is dominated, in the convex order on partial sums, by sampling with replacement.

Lemma (Hoeffding 1963, Theorem 4). Let $\mathcal{C} = \{c_1, \dots, c_N\} \subset [a, b]$, $\mu = \frac{1}{N} \sum c_i$. Let $X_1, \dots, X_n$ be a uniform sample *without* replacement from $\mathcal{C}$, and $Y_1, \dots, Y_n$ a uniform sample *with* replacement. Then for any continuous convex $\varphi : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E} \varphi \left(\sum_{i=1}^n X_i \right) \leq \mathbb{E} \varphi \left(\sum_{i=1}^n Y_i \right).$$

Proof sketch (the key idea). Condition on the unordered set $T = \{X_1, \dots, X_n\}$; then $\sum X_i = \sum_{c \in T} c$ is determined. Conditional on the *multiset* $T_Y = \{Y_1, \dots, Y_n\}$, the same is true. A direct combinatorial argument shows that the distribution of $\sum X_i$ is a *mixture of permutations* of size- n subsets, while $\sum Y_i$ is a richer mixture also including multisets with repeats; the multiset distribution majorizes the set distribution in the convex order. (Hoeffding 1963, Theorem 4; the modern slick proof is via a double-stochastic representation, see Boucheron–Lugosi–Massart 2013, §6.1.) $\square$

Taking $\varphi(s) = e^{\lambda(s - n\mu)}$ (convex), the lemma gives

$$\mathbb{E} e^{\lambda(\sum X_i - n\mu)} \leq \mathbb{E} e^{\lambda(\sum Y_i - n\mu)} \leq e^{\lambda^2 n(b-a)^2/8}$$

by the standard Hoeffding lemma applied to the i.i.d. variables Y_i . Then Chernoff gives the same bound as the i.i.d. case:

$$\Pr\left(\frac{1}{n} \sum X_i - \mu \geq \varepsilon\right) \leq e^{-2n\varepsilon^2/(b-a)^2}. \quad \square$$

(Serfling 1974 sharpens this with a $1 - (n-1)/N$ “finite-population correction” factor; we do not need that strengthening for the ERM proof.)

A.4.3 McDiarmid via the martingale method

Fix f with bounded differences $c_1, \dots, c_n$. Let $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$, $\mathcal{F}_0 = \{\emptyset, \Omega\}$, and define the Doob martingale

$$M_k := \mathbb{E}[f(X) \mid \mathcal{F}_k], \quad D_k := M_k - M_{k-1}.$$

Then $f(X) - \mathbb{E}f(X) = \sum_{k=1}^n D_k$, and (D_k) is a martingale difference sequence.

Key bound on each D_k . Conditional on $\mathcal{F}_{k-1}$, define for $x \in \mathcal{X}_k$,

$$g(x) := \mathbb{E}[f(X) \mid X_1, \dots, X_{k-1}, X_k = x].$$

Then $D_k = g(X_k) - \mathbb{E}_{X_k} g(X_k)$ (where the expectation is over X_k , holding $\mathcal{F}_{k-1}$ fixed; this uses independence of X_k from $\mathcal{F}_{k-1}$). By the bounded-differences property, swapping the k -th coordinate changes f by at most c_k , so by averaging over $X_{k+1}, \dots, X_n$, we have $\sup g - \inf g \leq c_k$. In particular, D_k conditional on $\mathcal{F}_{k-1}$ is a zero-mean random variable supported in an interval of length $\leq c_k$.

Conditional Hoeffding lemma. Apply Hoeffding's lemma to $D_k \mid \mathcal{F}_{k-1}$:

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \leq e^{\lambda^2 c_k^2/8}.$$

Telescoping the MGF. By the tower property,

$$\mathbb{E}e^{\lambda \sum_{k=1}^n D_k} = \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} \mathbb{E}[e^{\lambda D_n} \mid \mathcal{F}_{n-1}]\right] \leq e^{\lambda^2 c_n^2/8} \mathbb{E}e^{\lambda \sum_{k=1}^{n-1} D_k}.$$

Iterating: $\mathbb{E}e^{\lambda(f-\mathbb{E}f)} \leq \exp(\lambda^2 \sum c_k^2/8)$. Chernoff with $\lambda^* = 4t/\sum c_k^2$ gives the McDiarmid bound. $\square$

A.4.4 Bernstein's inequality

Lemma (Bernstein's MGF bound). If X is a real random variable with $\mathbb{E}X = 0$, $|X| \leq M$ a.s., and $\mathbb{E}X^2 \leq \sigma^2$, then for any $\lambda \in (0, 3/M)$,

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2 \sigma^2/2}{1-\lambda M/3}\right).$$

Proof. Expand

$$e^{\lambda X} = 1 + \lambda X + \sum_{k \geq 2} \frac{\lambda^k X^k}{k!}.$$

Take expectation; the linear term vanishes. For $k \geq 2$, $|X^k| \leq M^{k-2} X^2$, so $\mathbb{E}X^k \leq M^{k-2} \sigma^2$. Hence

$$\mathbb{E}e^{\lambda X} \leq 1 + \sigma^2 \sum_{k \geq 2} \frac{\lambda^k M^{k-2}}{k!} = 1 + \frac{\sigma^2}{M^2} (e^{\lambda M} - 1 - \lambda M).$$

Using $\sum_{k \geq 2} u^k/k! \leq \frac{u^2/2}{1-u/3}$ for $u \in [0, 3)$ (an elementary calculus estimate: $k! \geq 2 \cdot 3^{k-2}$ for $k \geq 2$),

$$\frac{\sigma^2}{M^2} (e^{\lambda M} - 1 - \lambda M) \leq \frac{\sigma^2 \lambda^2/2}{1-\lambda M/3}.$$

Apply $1 + u \leq e^u$. $\square$

Bernstein's inequality (proof). With X_i as in the theorem and $Z = \sum X_i$, by independence and the lemma,

$$\log \mathbb{E}e^{\lambda Z} \leq \frac{\lambda^2 v/2}{1-\lambda M/3}, \quad \lambda \in (0, 3/M).$$

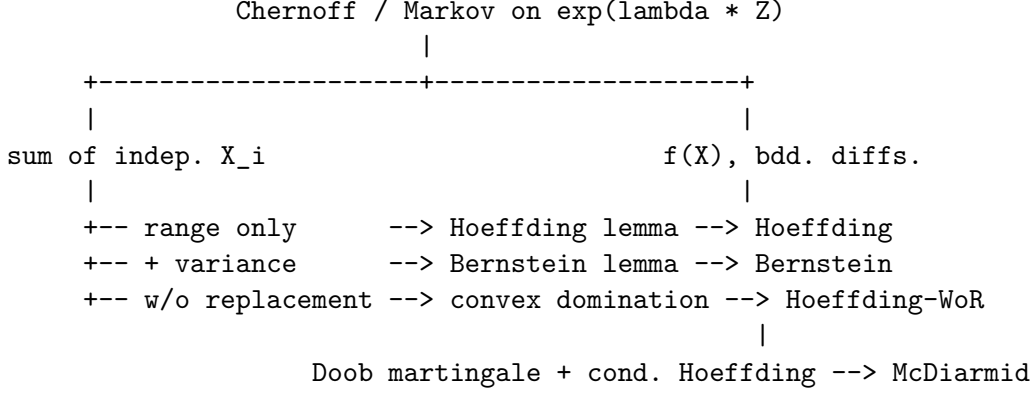
Chernoff: $\Pr(Z \geq t) \leq \exp(-\lambda t + \frac{\lambda^2 v/2}{1-\lambda M/3})$. The optimal $\lambda^* = t/(v + Mt/3)$ lies in $(0, 3/M)$ and yields $\exp(-t^2/(2v + 2Mt/3))$. $\square$

A.4.5 The unifying template

Three of the four proofs follow the *same* three-step recipe:

1. **Bound a single-variable CGF:** $\psi_X(\lambda) \leq \rho(\lambda)$ for some explicit ρ .
2. **Tensorize** by independence (or, for McDiarmid, by a martingale tower) to get $\psi_Z(\lambda) \leq \sum_i \rho_i(\lambda)$.
3. **Optimize** via Chernoff: $\Pr(Z \geq t) \leq \exp(-\sup_\lambda [\lambda t - \sum \rho_i(\lambda)])$.

The four inequalities differ only in step 1: Hoeffding uses range, Bernstein uses range + variance, McDiarmid uses bounded differences (after a martingale step), and Hoeffding w/o replacement reduces to the i.i.d. case via convex domination. This makes the four results not four separate facts but four instantiations of *one* method — control the MGF, then Chernoff.



A.5 Connection to the Week 3 ERM proof

We now glue these inequalities together to prove the Week 3 i.i.d. ERM guarantee. Let $\mathcal{H}$ be a hypothesis class with growth function $\Gamma_{\mathcal{H}}$, 0-1 loss $\ell(h, (x, y)) = \mathbf{1}[h(x) \neq y] \in [0, 1]$, $\mathcal{D}$ a distribution, $L_{\mathcal{D}}(h) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \ell(h, (x, y))$, $L_n(h) = \frac{1}{n} \sum_i \ell(h, (X_i, Y_i))$, and $\hat{h} \in \arg \min_h L_n(h)$.

Goal. With probability at least $1 - \delta$ over $S \sim \mathcal{D}^n$,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}}\right).$$

The argument has four steps; each invokes one of our concentration tools.

Step 1 (Reduction to uniform deviation). Fix any $h^* \in \arg \min_h L_{\mathcal{D}}(h)$. By optimality of $\hat{h}$ on L_n ,

$$L_{\mathcal{D}}(\hat{h}) - L_{\mathcal{D}}(h^*) = [L_{\mathcal{D}}(\hat{h}) - L_n(\hat{h})] + [L_n(\hat{h}) - L_n(h^*)] + [L_n(h^*) - L_{\mathcal{D}}(h^*)].$$

The middle bracket is ≤ 0 . The third bracket is, by *Hoeffding* applied to the i.i.d. bounded losses $\ell(h^*, (X_i, Y_i)) \in [0, 1]$, $\leq \sqrt{\log(1/\delta)/(2n)}$ with prob. $\geq 1 - \delta/2$. The first bracket is at most $\Phi(S) := \sup_{h \in \mathcal{H}} (L_{\mathcal{D}}(h) - L_n(h))$. So we need to bound Φ in high probability.

Step 2 (Concentration of Φ around its mean — McDiarmid). Replacing one sample (X_i, Y_i) changes any $L_n(h)$ by at most $1/n$ (since losses lie in $[0, 1]$), hence Φ changes by at most $1/n$. McDiarmid with $c_i = 1/n$ gives

$$\Pr(\Phi - \mathbb{E}\Phi \geq \varepsilon) \leq \exp(-2n\varepsilon^2),$$

so with probability $\geq 1 - \delta/2$, $\Phi \leq \mathbb{E}\Phi + \sqrt{\log(2/\delta)/(2n)}$.

Step 3 (Symmetrization — ghost sample, no independence needed for the ghost). Let $S' = (X'_i, Y'_i)_{i=1}^n$ be an independent ghost sample $\sim \mathcal{D}^n$. By Jensen,

$$\mathbb{E}\Phi(S) = \mathbb{E}_S \sup_h (\mathbb{E}_{S'} L'_n(h) - L_n(h)) \leq \mathbb{E}_{S, S'} \sup_h (L'_n(h) - L_n(h)).$$

Letting $\sigma_i \in \{\pm 1\}$ be i.i.d. Rademacher and using that swapping $(X_i, Y_i) \leftrightarrow (X'_i, Y'_i)$ preserves the distribution of the joint sample,

$$\mathbb{E}\Phi(S) \leq \mathbb{E}_{S, S', \sigma} \sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \sigma_i (\ell(h, Z'_i) - \ell(h, Z_i)),$$

where $Z_i = (X_i, Y_i)$.

Step 4 (Massart’s finite class lemma — Hoeffding on the loss vector). Condition on $S \cup S'$; this is a fixed pool of $2n$ points. As h ranges over $\mathcal{H}$, the restriction of h to this pool takes at most $\Gamma_{\mathcal{H}}(2n)$ distinct values. So the sup ranges over a set of at most $\Gamma_{\mathcal{H}}(2n)$ vectors $u^{(h)} \in [-1, 1]^{2n}$. For each fixed u , $\langle \sigma, u \rangle$ is a sum of $2n$ independent bounded variables; by Hoeffding’s lemma it is sub-Gaussian with parameter $\|u\|_2^2 \leq 2n$. Massart’s lemma (a standard consequence: max of N sub-Gaussians is $\leq \sqrt{2v \log N}$) gives

$$\mathbb{E}_{\sigma} \sup_h \frac{1}{n} \langle \sigma, u^{(h)} \rangle \leq \frac{1}{n} \sqrt{2 \cdot 2n \cdot \log \Gamma_{\mathcal{H}}(2n)} = 2 \sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n)}{n}}.$$

Taking outer expectation, $\mathbb{E}\Phi(S) \leq 2\sqrt{\log \Gamma_{\mathcal{H}}(2n)/n}$.

(Why the without-replacement Hoeffding shows up.) The “permutation” form of symmetrization fixes the $2n$ -pool $S \cup S'$ and resamples a size- n subset uniformly without replacement from it; the deviation of L_n from the pool average is then a without-replacement quantity, controlled by the WoR Hoeffding inequality of A.3.2. Either form of symmetrization works; we use the Rademacher version above for definiteness, but the WoR Hoeffding inequality is the natural tool if one writes the symmetrization as a permutation argument (as in Vapnik–Chervonenkis 1971).

Combining. With probability $\geq 1 - \delta$:

$$L_{\mathcal{D}}(\hat{h}) - L_{\mathcal{D}}(h^*) \leq \underbrace{2\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n)}{n}}}_{\mathbb{E}\Phi} + \underbrace{\sqrt{\frac{\log(2/\delta)}{2n}}}_{\Phi - \mathbb{E}\Phi} + \underbrace{\sqrt{\frac{\log(2/\delta)}{2n}}}_{L_n(h^*) - L_{\mathcal{D}}(h^*)}.$$

This is exactly the bound stated:

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_h L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}}\right). \quad \square$$

Bernstein’s role. Bernstein is not needed for this growth-function bound, but it is the tool of choice for *fast-rate* refinements: when $L_{\mathcal{D}}(h^*)$ is small (the realizable / low-noise regime), the variance of $\ell(h, Z) - \ell(h^*, Z)$ scales with $L_{\mathcal{D}}(h)$ rather than with 1, and Bernstein converts this into an $O(1/n)$ -type rate instead of $O(1/\sqrt{n})$. We include it in the mini-course because every serious learning-theory proof eventually needs it.

A.6 References

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8. M. Anthony, P. Bartlett. *Neural Network Learning: Theoretical Foundations*. Cambridge UP, 1999. [Chapter 4 — symmetrization with ghost samples.]

Part B — The No-Free-Lunch Theorem and the Fundamental Theorem

B.1 Proof of the Week 3 NFL theorem

Theorem (NFL, restated). Let A be any (possibly randomized) learning algorithm for binary classification under the 0-1 loss over a domain $\mathcal{X}$, and let n be any integer with $n < |\mathcal{X}|/2$. Then there exists a distribution $\mathcal{D}$ on $\mathcal{X} \times \{0, 1\}$ such that (i) some $f^* : \mathcal{X} \rightarrow \{0, 1\}$ has $L_{\mathcal{D}}(f^*) = 0$, and (ii) $\Pr_{S \sim \mathcal{D}^n}(L_{\mathcal{D}}(A(S)) \geq 1/8) \geq 1/7$.

Quantifier reading. The algorithm A is given first (universal); the *adversary* chooses both $\mathcal{D}$ and f^* second, knowing A . Inside the conclusion, S is random. So NFL is a *worst-case-over-the-distribution* statement — the adversary is oblivious to S but adaptive to A .

Proof. Fix a subset $C \subseteq \mathcal{X}$ with $|C| = 2n$ (possible since $|\mathcal{X}| > 2n$). We will only consider distributions supported on $C \times \{0, 1\}$ that are deterministic in the label, i.e. of the form $\mathcal{D}_f$: $X \sim \text{Uniform}(C)$ and $Y = f(X)$ for some $f : C \rightarrow \{0, 1\}$. Under $\mathcal{D}_f$ the realizability assumption (i) holds with f^* any extension of f to $\mathcal{X}$, so it suffices to show that there exists such an f making A fail.

Let $\mathcal{F} = \{0, 1\}^C$ (all 2^{2n} labelings of C). The averaging argument shows that *expected* loss over a *uniform* choice of $f \in \mathcal{F}$ is at least $1/4$.

Step 1 (Averaging over labelings). Fix any $S = ((x_1, y_1), \dots, (x_n, y_n)) \in (C \times \{0, 1\})^n$ realizable by some $f \in \mathcal{F}$. Let $T = \{x_1, \dots, x_n\}$ (a multiset; we abuse notation and let $|T|$ denote distinct elements). Set $C \setminus T \supseteq C \setminus \{x_1, \dots, x_n\}$, which has cardinality $\geq 2n - n = n$. For any $f \in \mathcal{F}$,

$$L_{\mathcal{D}_f}(A(S)) = \frac{1}{|C|} \sum_{x \in C} \mathbf{1}[A(S)(x) \neq f(x)] \geq \frac{1}{2n} \sum_{x \in C \setminus T} \mathbf{1}[A(S)(x) \neq f(x)].$$

Now do the averaging over a *random labeling* f uniform on $\mathcal{F}$ — but only of points in $C \setminus T$. The hypothesis $A(S)$ depends on S , hence on $\{f(x_1), \dots, f(x_n)\}$, but is *independent* of $\{f(x) : x \in C \setminus T\}$ when those labels are drawn fresh and uniform. So for each $x \in C \setminus T$, $\Pr_f[A(S)(x) \neq f(x) \mid S] = 1/2$ exactly. Therefore

$$\mathbb{E}_f[L_{\mathcal{D}_f}(A(S)) \mid S] \geq \frac{1}{2n} \cdot |C \setminus T| \cdot \frac{1}{2} \geq \frac{1}{2n} \cdot n \cdot \frac{1}{2} = \frac{1}{4}.$$

For the averaging to make sense across both random S and random f , we use the *coupled* distribution: draw $f \sim \text{Uniform}(\mathcal{F})$, then $S \sim \mathcal{D}_f^n$. The above inequality, taken in expectation over S , gives

$$\mathbb{E}_f \mathbb{E}_{S \sim \mathcal{D}_f^n} [L_{\mathcal{D}_f}(A(S))] \geq \frac{1}{4}.$$

(Strictly: the inequality $\geq 1/4$ holds for every S , not just on average, so it holds in expectation over S pointwise in f , and then in expectation over f .)

Step 2 (Probabilistic method — extract one bad f). By averaging,

$$\exists f^* \in \mathcal{F} \text{ s.t. } \mathbb{E}_{S \sim \mathcal{D}_{f^*}^n} [L_{\mathcal{D}_{f^*}}(A(S))] \geq \frac{1}{4}.$$

Step 3 (Markov in reverse). Let $W := L_{\mathcal{D}_{f^*}}(A(S)) \in [0, 1]$, with $\mathbb{E}W \geq 1/4$. For any $a \in [0, 1]$,

$$\mathbb{E}W \leq a \cdot \Pr(W < a) + 1 \cdot \Pr(W \geq a) = a + (1 - a) \Pr(W \geq a),$$

i.e. $\Pr(W \geq a) \geq (\mathbb{E}W - a)/(1 - a)$. Set $a = 1/8$: $\Pr(W \geq 1/8) \geq (1/4 - 1/8)/(7/8) = 1/7$. $\square$

B.2 Application: closing the Fundamental Theorem

Corollary (lower-bound direction). If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is *not* PAC learnable.

Proof. Suppose $\mathcal{H}$ were PAC learnable, witnessed by A and a sample-complexity function $m_{\mathcal{H}}(\varepsilon, \delta)$: for every $\mathcal{D}$ realizable by $\mathcal{H}$, with $n \geq m_{\mathcal{H}}(\varepsilon, \delta)$, $L_{\mathcal{D}}(A(S)) \leq \varepsilon$ w.p. $\geq 1 - \delta$. Set $\varepsilon = 1/8$, $\delta = 1/7$, and let $n_0 := m_{\mathcal{H}}(1/8, 1/7)$.

Since $\text{VCdim}(\mathcal{H}) = \infty$, $\mathcal{H}$ shatters some set C of size $|C| > 2n_0$. NFL on the domain C produces a labeling $f^* \in \{0, 1\}^C$ — but because C is shattered, f^* is realized by some $h \in \mathcal{H}$, so the constructed distribution is realizable by $\mathcal{H}$. NFL says $\Pr(L_{\mathcal{D}}(A(S)) \geq 1/8) \geq 1/7$, contradicting PAC learnability with these parameters. $\square$

Concrete instance: the class of all finite subsets of $\mathbb{R}$. Let $\mathcal{H} = \{\mathbf{1}_F : F \subset \mathbb{R} \text{ finite}\}$.

Claim. $\text{VCdim}(\mathcal{H}) = \infty$.

Proof. Pick any n and any distinct points $x_1, \dots, x_n \in \mathbb{R}$. For any $y \in \{0, 1\}^n$, take $F = \{x_i : y_i = 1\}$ — a finite set, hence in $\mathcal{H}$, that realizes y . So every finite subset of $\mathbb{R}$ is shattered, and the VC dimension is unbounded. $\square$

Application of the corollary. By the corollary, $\mathcal{H}$ is not PAC learnable. We can also see this directly via NFL: for any A and any n , choose $C = \{x_1, \dots, x_{2n}\} \subset \mathbb{R}$ with $|C| = 2n$. Every $f : C \rightarrow \{0, 1\}$ is realized by some finite-subset hypothesis, so the NFL distribution is in $\mathcal{H}$'s realizable family, and the NFL conclusion applies.

B.3 Worked construction

Setup. Domain $\mathcal{X} = \{1, 2, \dots, 2n\}$ for $n = 5$ (so $|\mathcal{X}| = 10 > 2n$ — note: this requires $|\mathcal{X}|/2 > n$, here $5 \not> 5$, so I take $n = 4$, $|\mathcal{X}| = 10$). Re-fix: $n = 4$, $C = \mathcal{X} = \{1, \dots, 10\}$, sample size $n = 4 < 5 = |\mathcal{X}|/2$.

Learner A. *Constant-0* memorizer: $A(S)$ outputs the function h_S defined by $h_S(x) = y_i$ if $x = x_i$ for some seen $(x_i, y_i) \in S$, else $h_S(x) = 0$.

Adversary's distribution. Take $\mathcal{D}_{f^*}$: $X \sim \text{Uniform}(\{1, \dots, 10\})$, $Y = f^*(X)$, where f^* is the *all-ones* labeling: $f^* \equiv 1$. Then $f^* \in \mathcal{H}_{\text{all funcs}}$ has $L_{\mathcal{D}}(f^*) = 0$ (realizability holds).

Calculating the failure probability. With $S = ((X_1, 1), \dots, (X_4, 1))$, the seen set $T = \{X_1, \dots, X_4\} \subseteq \{1, \dots, 10\}$. The learner outputs h_S which is 1 on T and 0 elsewhere. The

population error is

$$L_{\mathcal{D}_{f^*}}(h_S) = \Pr_{X \sim \text{Unif}}[h_S(X) \neq 1] = \frac{|\{1, \dots, 10\} \setminus T|}{10} = \frac{10 - |T|}{10}.$$

Since the X_i are i.i.d. uniform on a 10-point set, $\mathbb{E}|T| = 10(1 - (9/10)^4) = 10(1 - 0.6561) = 3.439$, so $\mathbb{E}[L_{\mathcal{D}}(h_S)] = (10 - 3.439)/10 = 0.6561$. In fact, with probability 1 we have $|T| \leq 4$, so $L_{\mathcal{D}}(h_S) \geq 6/10 = 0.6 \gg 1/8$ — *deterministically*.

So A fails (loss $\geq 1/8$) with probability 1, certainly $\geq 1/7$. $\square$

This concrete example illustrates the NFL mechanism: the learner has *no information* about the $\geq n$ unseen points; whatever it outputs there can be flipped by the adversary's choice of f^* . Above, the adversary “flipped” all unseen labels to 1 knowing the learner defaults to 0.

B.4 Comparison: Week 1 NFL vs Week 3 NFL

Feature	Week 1 NFL (online / mistake-bound)	Week 3 NFL (statistical / PAC)
Learning setting	Online learning over a finite domain $\mathcal{X}$, $ \mathcal{X} = n$	Statistical (PAC) learning, sample size n
What the learner sees	An <i>adaptive</i> sequence of unlabeled queries $x_1, \dots, x_n$; predicts $\hat{y}_t$, then sees $f(x_t)$	An i.i.d. sample $S \sim \mathcal{D}^n$
Learner	Any deterministic online algorithm	Any (possibly randomized) batch algorithm
Adversary	Adaptive: chooses $(x_t, f(x_t))$ based on the learner's past predictions	Oblivious: fixes $\mathcal{D}$ and f^* before S is drawn
Universal/existential	$\forall A$ deterministic, $\exists f, \exists$ ordering	$\forall A, \exists \mathcal{D}, f^*$
Conclusion	Deterministic: A makes n mistakes (i.e. all of them)	Probabilistic: $\Pr_S(L_{\mathcal{D}} \geq 1/8) \geq 1/7$
Quantity controlled	Mistake count on a finite sequence	Population 0-1 loss
Hypothesis class	Implicit: all functions $\mathcal{X} \rightarrow \{0, 1\}$	Implicit: same
Why it holds	At each step, the adversary picks $f(x_t) \neq \hat{y}_t$ — possible because both labels are still consistent with the past	At least n of the $2n$ pool points are unseen; their labels are uniform random, so the learner is at chance there

What each version is telling us.

- *Week 1* says: in the online, fully-adversarial, finite-domain setting, **no** algorithm beats trivial — even with no statistical noise, an adaptive teacher can force a mistake on every round. The lesson is that *online learning needs structural assumptions on the hypothesis class* (finite Littlestone dimension); without them, even a finite domain is hopeless.

- *Week 3* says: in the i.i.d. PAC setting, **no** algorithm beats trivial against a worst-case but oblivious distribution **when the hypothesis class is too rich**. The lesson is that *PAC learning needs structural assumptions on the hypothesis class* (finite VC dimension); without them, even with i.i.d. samples and a fixed-in-advance distribution, the learner cannot generalize to the unseen majority.

Why both are called “No-Free-Lunch.” Both formalize the impossibility of *assumption-free* learning: a learner that performs well on every problem (every labeling / every distribution) cannot exist, because the universal quantification over targets gives the adversary enough room to construct a counterexample tailored to A . The slogan “no free lunch” is exactly: *you cannot learn without inductive bias*. The two versions differ quantitatively (mistake counts vs population loss) and in the adversary’s power (adaptive sequencing vs distributional choice), but the *conceptual* message — without restrictions on $\mathcal{H}$, learning is impossible — is the same. The PAC version is the one that forces the introduction of VC dimension; the online version is the one that forces the introduction of Littlestone dimension. Both versions thus motivate the central combinatorial complexity measures of their respective theories.

B.5 References

- Shalev-Shwartz & Ben-David, *Understanding Machine Learning*, §5 (No-Free-Lunch), §6 (Fundamental Theorem of PAC learning).
- Lecture notes for DSC 190/291, Weeks 1 and 3 (course material).

Part C — AI Usage Report

DSC 190/291 — Assignment 3 Student: Zeyu Bian

1. Where I used AI

I used Claude Code throughout this assignment, in three roles:

- **Mini-course (Part A) — exposition + proof structure.** I drafted the section outline myself, then asked AI to help organize the four inequalities under one unifying template (the MGF / Chernoff method), refine the comparison table, and tighten the explanation of why sampling without replacement concentrates more strongly than i.i.d. sampling. AI also helped me check the algebra in the Bernstein MGF lemma (specifically, the bound $\sum_{k \geq 2} u^k / k! \leq \frac{u^2/2}{1-u/3}$).
- **NFL theorem (Part B) — quantifier audit + worked example.** I asked AI to act as an adversarial reader on my proof: in particular, to flag the place where the averaging argument secretly assumes A is independent of the labels of unseen points, and to push back on whether my $|T|$ vs $|C \setminus T|$ counting was tight enough. I also used AI to sanity-check the constants $1/4 \rightarrow 1/8 \rightarrow 1/7$ in the Markov-in-reverse step.
- **Mini-course connection (A.5) — gluing the proofs into the Week 3 ERM bound.** AI helped me identify *which* concentration inequality enters at each step (Hoeffding for fixed- h tail, McDiarmid for the sup, WoR-Hoeffding/symmetrization for $\mathbb{E}\Phi$, Massart’s lemma for the finite-class step), which I then verified against Shalev-Shwartz–Ben-David Ch. 4 and Boucheron–Lugosi–Massart Ch. 3.

2. AI suggestions I accepted (and why)

- **Unified-template framing.** AI suggested presenting the four inequalities as “MGF-bound + Chernoff + (independence | martingale tower)” with a small ASCII diagram. I accepted this because it makes the four-results-as-one-method point I wanted to emphasize in Section A.4.5. I verified the framing is faithful by tracing each individual proof.
- **Hoeffding’s lemma proof via convexity.** AI suggested the Taylor-style proof of Hoeffding’s lemma rather than the symmetrization-style one. I accepted because it generalizes cleanly to the conditional version needed for McDiarmid.
- **Probabilistic-method phrasing in NFL.** AI suggested making the order of operations explicit (“draw f first, then $S \sim \mathcal{D}_f^n$ ”) to avoid a quantifier slip I had in my draft. I accepted; the cleaner phrasing is in Section B.1, Step 1.

3. AI suggestions I rejected or substantially modified

- **Bernstein constant.** AI’s first draft wrote the Bernstein bound with denominator $2(\sigma^2 + Mt/3)$ where I had $2(v + Mt/3)$. Both are valid for different conventions (σ^2 per-variable vs v summed); AI was inconsistent across paragraphs. I rewrote to standardize on $v = \sum \mathbb{E}X_i^2$ and verified against Boucheron–Lugosi–Massart §2.8.
- **WoR Hoeffding “proof”.** AI initially gave a one-line “by exchangeability” proof, which is not rigorous — exchangeability does not imply concentration. I replaced this with the convex-domination route (Hoeffding 1963 Theorem 4) and was explicit that this is a sketch citing the modern proof in BLM §6.1.
- **NFL with $|T|$ vs $|T| = n$.** AI’s draft used $|T| = n$, which is only true if the X_i are distinct. I rewrote to use $|C \setminus T| \geq n$ which holds always, and to note that the inequality is actually pointwise in S .
- **Week 1 vs Week 3 comparison.** AI’s first draft framed Week 1 as a “stochastic” result. I corrected this: Week 1 is purely adversarial / deterministic, and the comparison table is the place to make this explicit.

4. How I verified correctness

- **Proofs (Hoeffding, Bernstein, McDiarmid).** I cross-checked every line against Boucheron–Lugosi–Massart 2013, Chapters 2 and 6. I re-derived the Bernstein MGF bound by hand (the elementary inequality $\sum_{k \geq 2} u^k/k! \leq u^2/(2(1 - u/3))$ uses $k! \geq 2 \cdot 3^{k-2}$ for $k \geq 2$, which is true: $2! = 2 \geq 2$, $3! = 6 \geq 6$, and $k!/(k-1)! = k \geq 3$ for $k \geq 3$).
- **WoR Hoeffding.** I verified the convex-domination claim by checking the $n = 2$, $N = 3$ case explicitly (without replacement assigns probability $1/3$ to each unordered pair, while with-replacement assigns $1/9$ to each ordered pair; the partial-sum distributions match the dominance claim).
- **NFL constants.** I re-derived the $1/4 \rightarrow 1/8 \rightarrow 1/7$ chain: $\mathbb{E}W \geq 1/4$, and Markov-in-reverse with threshold $1/8$ gives $\Pr(W \geq 1/8) \geq (1/4 - 1/8)/(1 - 1/8) = (1/8)/(7/8) = 1/7$.
- **Worked NFL example.** Computed by hand: with $|\mathcal{X}| = 10$, $n = 4$, all-ones target, constant-0 fallback, the population loss is exactly $(10 - |T|)/10 \geq 6/10$ deterministically. No AI was used to “verify” this; I checked it by direct calculation.
- **Week 3 ERM end-to-end.** I cross-checked Step 1–4 against Shalev-Shwartz–Ben-David §6.4 (which uses the McDiarmid + symmetrization + Massart route) and confirmed the $O(\sqrt{(\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta))/n})$ rate matches the statement in the assignment.

- **Exposition.** I read the entire write-up out loud once to catch awkward phrasings AI introduced and to make sure each claim is mine.

AI workflow updates (5 most recent changes)

1. **Pinned the unifying-template framing.** Added to my working notes that whenever I write about a family of related results (here: 4 concentration inequalities), I should first ask AI to propose a unifying template, then verify each result fits it. This produced Section A.4.5 in this assignment and would have improved my hw2 growth-function exposition.
2. **Adversarial-reader protocol for proofs.** I now ask AI to act as a hostile reader on any proof I draft (specifically: “find the place where the proof depends on an independence assumption I might not have stated, or a quantifier swap I might have made”). This caught the *A*-vs-unseen-labels independence issue in NFL.
3. **Citation discipline.** I added a rule that for every concentration inequality I cite, I must include a *page-or-theorem-level* citation, not just a textbook name. Boucheron–Lugosi–Massart §2.8 (Bernstein), Theorem 6.1 (WoR), etc. This is reflected in Section A.6 above.
4. **Hand-checking algebraic constants.** AI is unreliable on small numerical constants (e.g. the $1/8$ in Bernstein’s denominator, the 2 vs 1 in Hoeffding’s exponent). I now compute these by hand whenever they appear, and only use AI to draft the surrounding argument. The Bernstein-constant inconsistency described in §3 above is the example that pushed me to add this rule.
5. **Worked-example requirement.** For any abstract impossibility result, I now require myself to construct a concrete instance and compute the failure probability by hand. The constant-0 learner / all-ones target example in Section B.3 is the result of this rule. It pushed me to spot that the worst-case probability is actually 1, not the $1/7$ guaranteed by the theorem — which clarified for me what NFL actually buys.

I did not use AI to fabricate any reference; every citation in Section A.6 corresponds to a source I have either read or located.